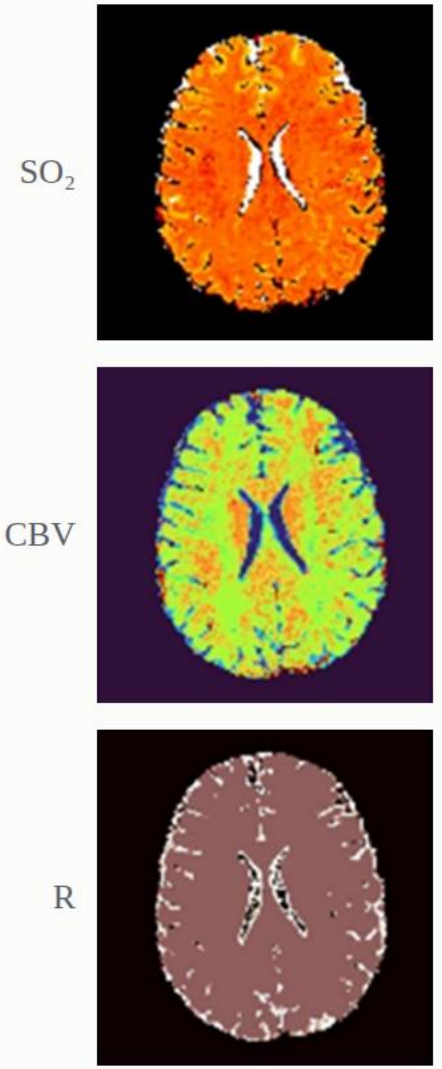
Technical Review of Magnetic Resonance Fingerprinting Applications in Cerebral Physiology

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Journal club · Andres Michel
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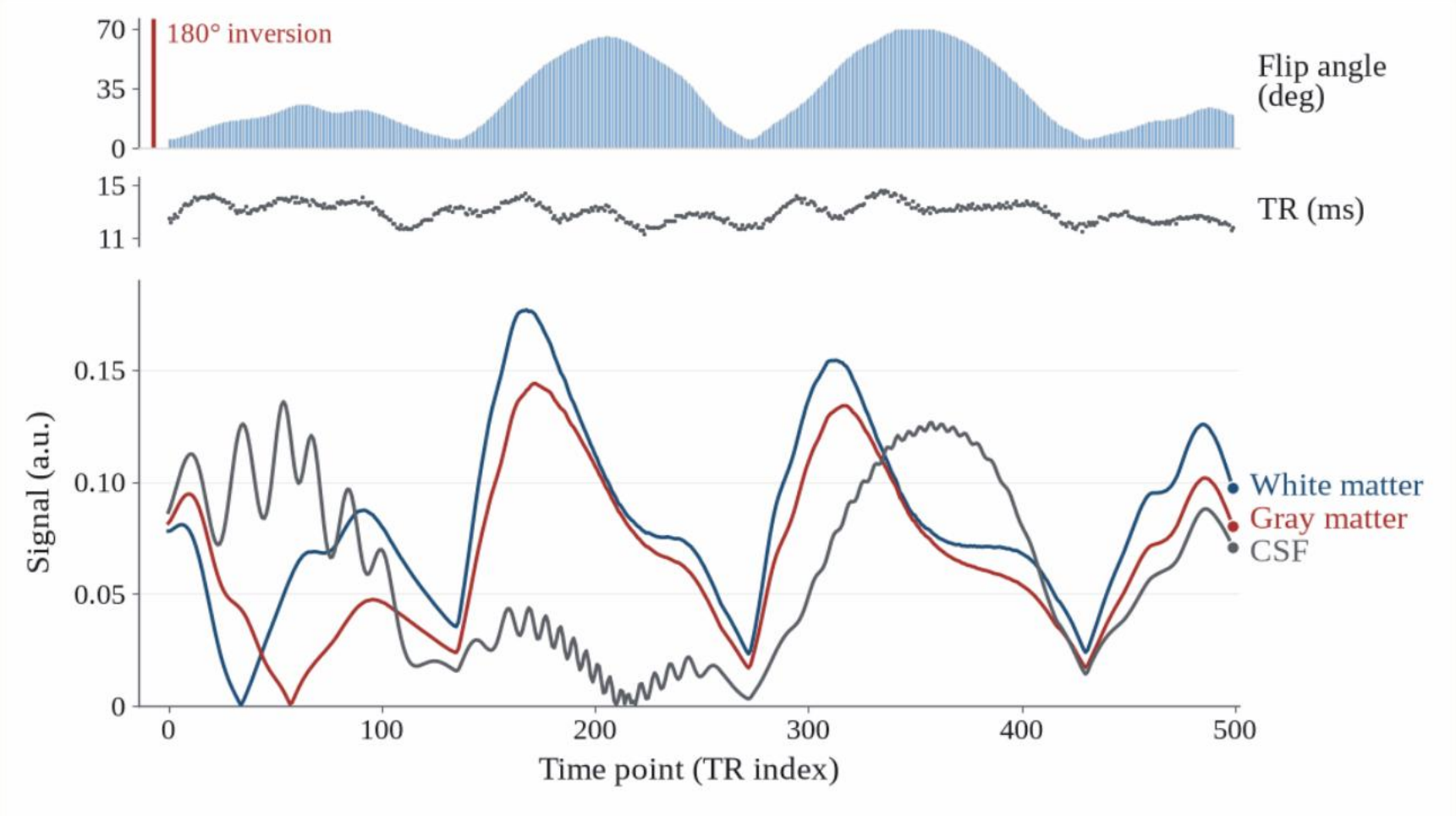
Standard methods each cover one part of the picture

Technique	What it gives	Trade-off
DSC MRI	CBV, CBF, mean transit time	Gadolinium bolus; limited in renal impairment
ASL	CBF	No contrast, but low SNR and resolution
DCE MRI	Ktrans (blood–brain barrier leakage)	Gadolinium; T1-weighted dynamic scan

Several scans: a longer exam, possible misregistration, and data that are hard to combine.

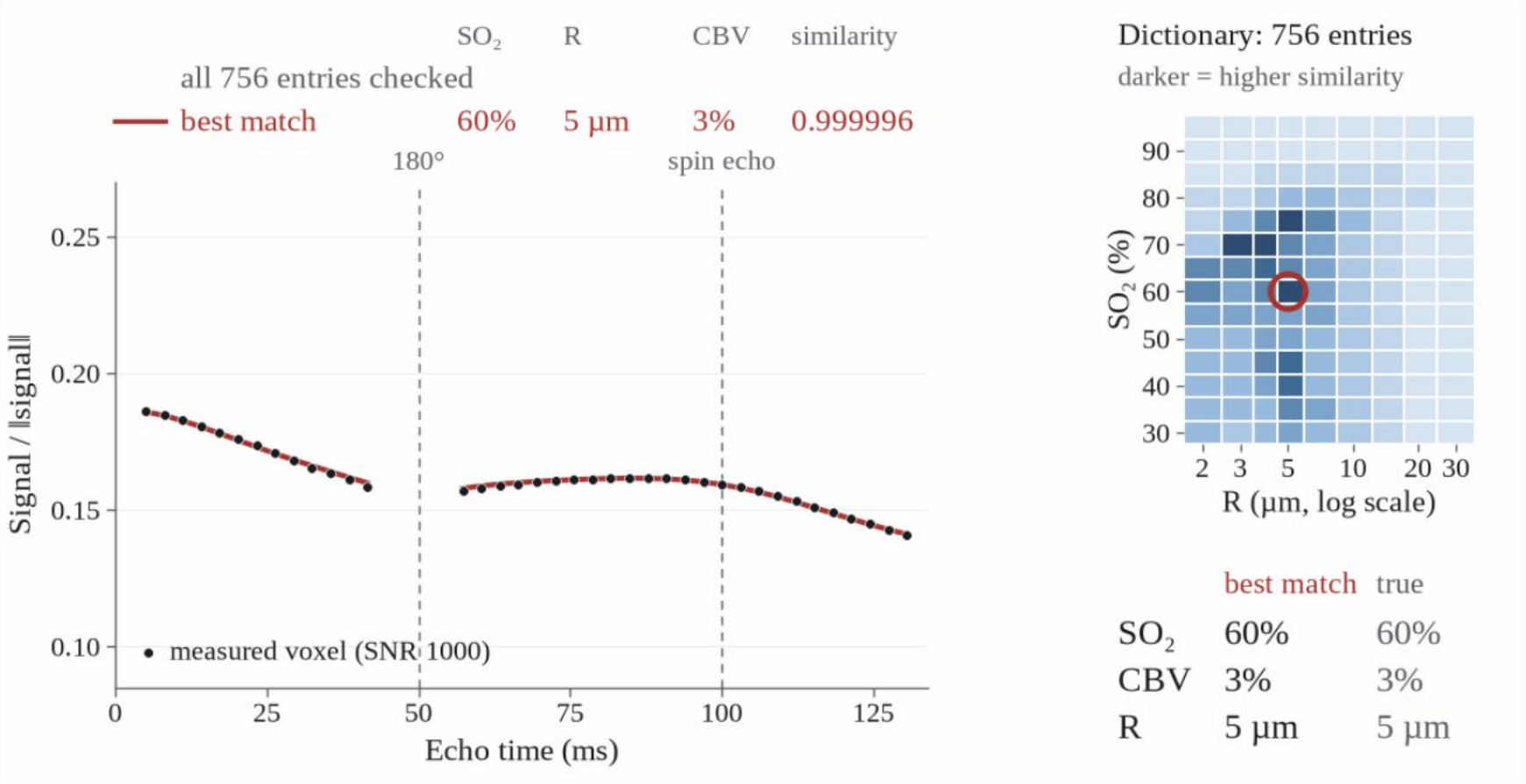
Physiological MRF	CBF, BAT, CBV, vessel radius, SO_2, water exchange, plus T1, T2, B1+	Each variant gets several of these from one acquisition and one model

MRF lets the signal vary on purpose



Concept: Ma et al., Nature 2013 (ref. 14 in Lin et al.) · Animation: simulated inversion-recovery FISP train, 3 T tissue values

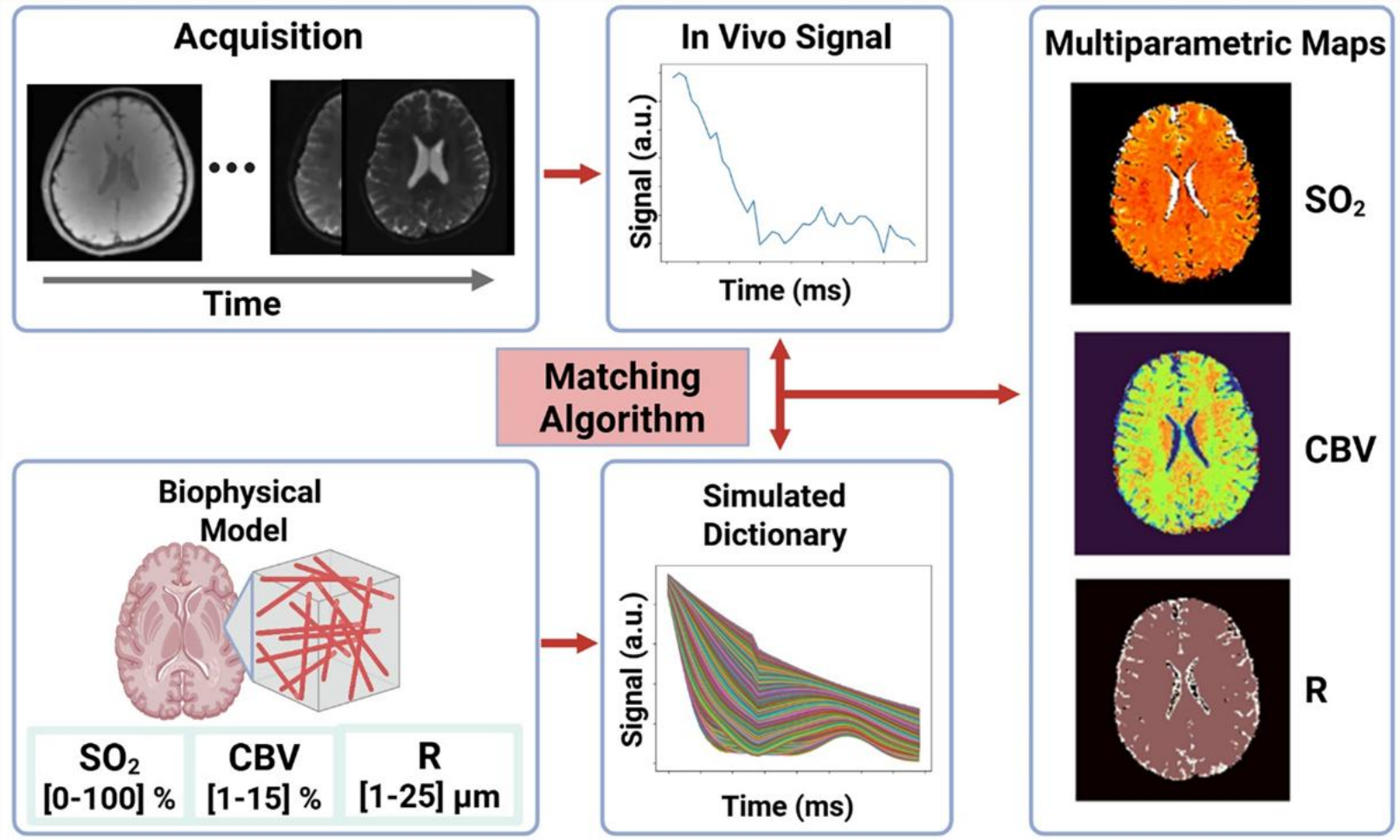
Reading a fingerprint: dictionary matching



Normalize every curve → inner product with each entry → keep the highest → that entry gives SO_2, CBV and R

Animation: small MRvF dictionary simulated for this talk (2D Monte Carlo, 38-echo GESFIDE timing)

Physiological MRF puts physiology into the dictionary



Lin, Lu & Fan, Magn Reson Med 2025, Fig. 1

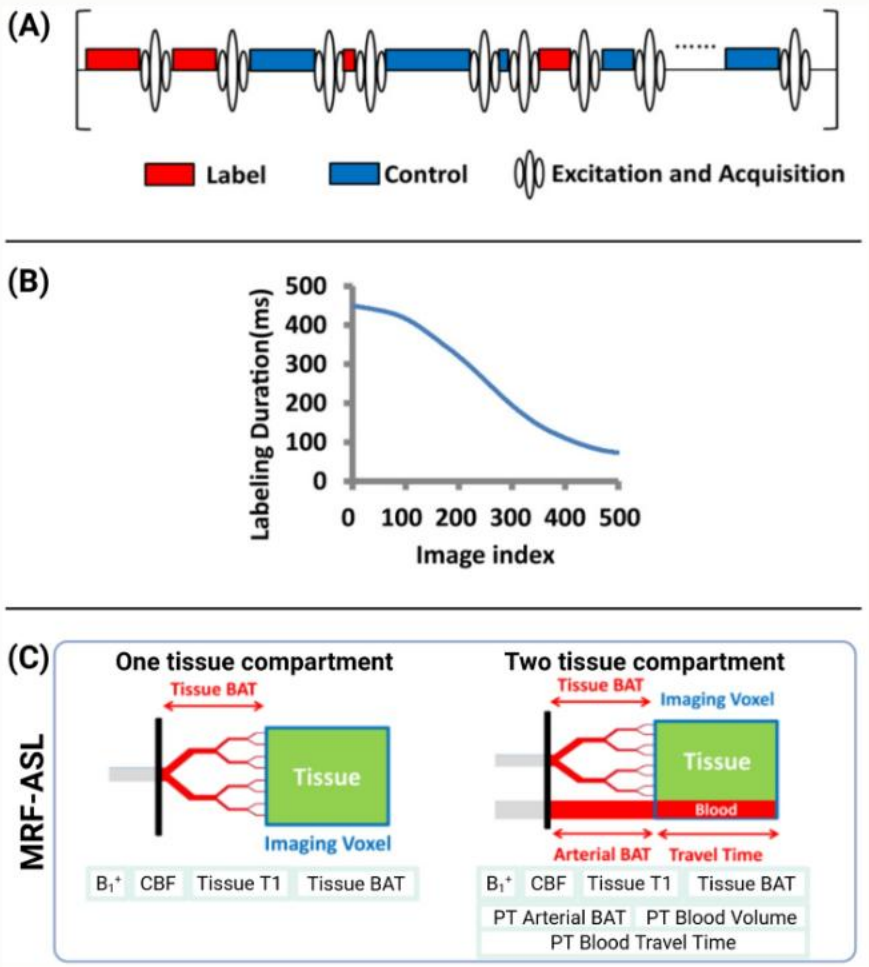
The review sorts the field into three families

	MRF-ASL	MRvF	MRF-VFD
Physiology	Blood flow and transit	Vessel structure and oxygenation	Vessel function and permeability
Main outputs	CBF, bolus arrival time, tissue T1, B1+	CBV, mean vessel radius R, SO_2; CVR when repeated	Ktrans, kep, water residence time τb, CBV
Signal encoded by	Pseudo-random label and control blocks, varying label durations	Multi-echo gradient and spin echoes (GESFIDE, SAGE) or bSSFP	Contrast dynamics (DCE, DSC) or non-contrast T1-based MRF
Dictionary model	One- or two-compartment kinetic model	Monte Carlo virtual voxel of vessels	Blood and tissue compartments exchanging water or contrast

Many of these methods also return supporting parameters such as T1, T2 and B1+ (Table 1).

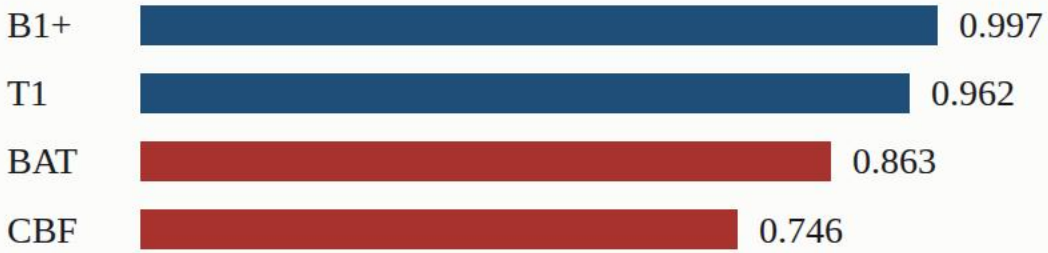
Lin, Lu & Fan, Magn Reson Med 2025, Tables 1 and 2

MRF-ASL: randomize the labeling



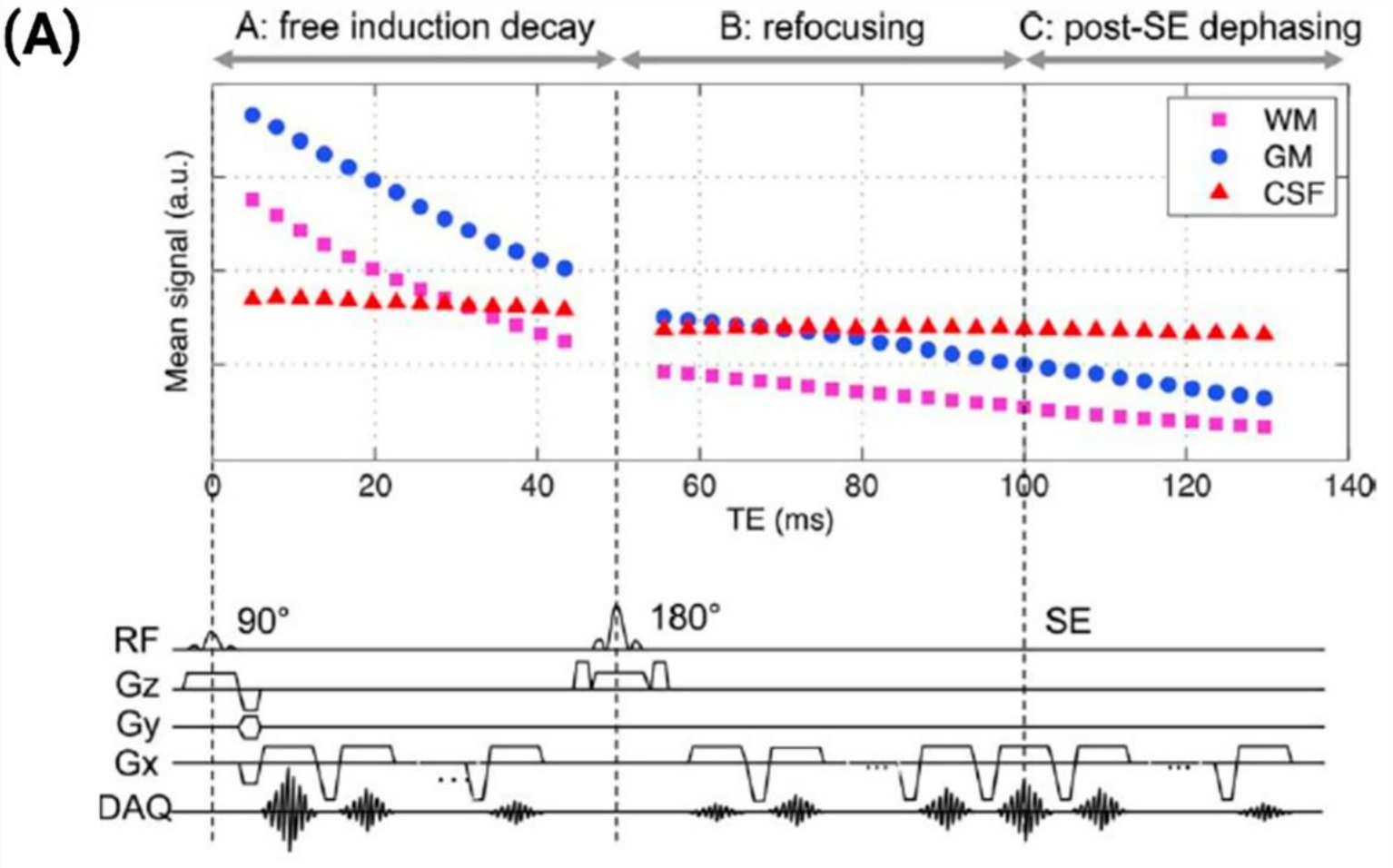
- (A, B) Label and control blocks in pseudo-random order, with labeling durations that shrink over the scan: hundreds of effective delays in one scan
- (C) One compartment: B1+, CBF, tissue T1, tissue BAT. Two compartments add arterial BAT, blood volume and travel time
- Wright 2018: CBF, arterial and tissue BAT in 2.5 min; CBF agreed with pCASL

Test–retest spatial correlation, 20 ± 19 days apart (Su et al., NMR Biomed 2020)



Red: parameters that depend on the kinetic model and the small (~1%) label signal

MRvF: the GESFIDE echo train



A Gradient echoes after the 90°. The signal falls with R2*, which includes static dephasing.

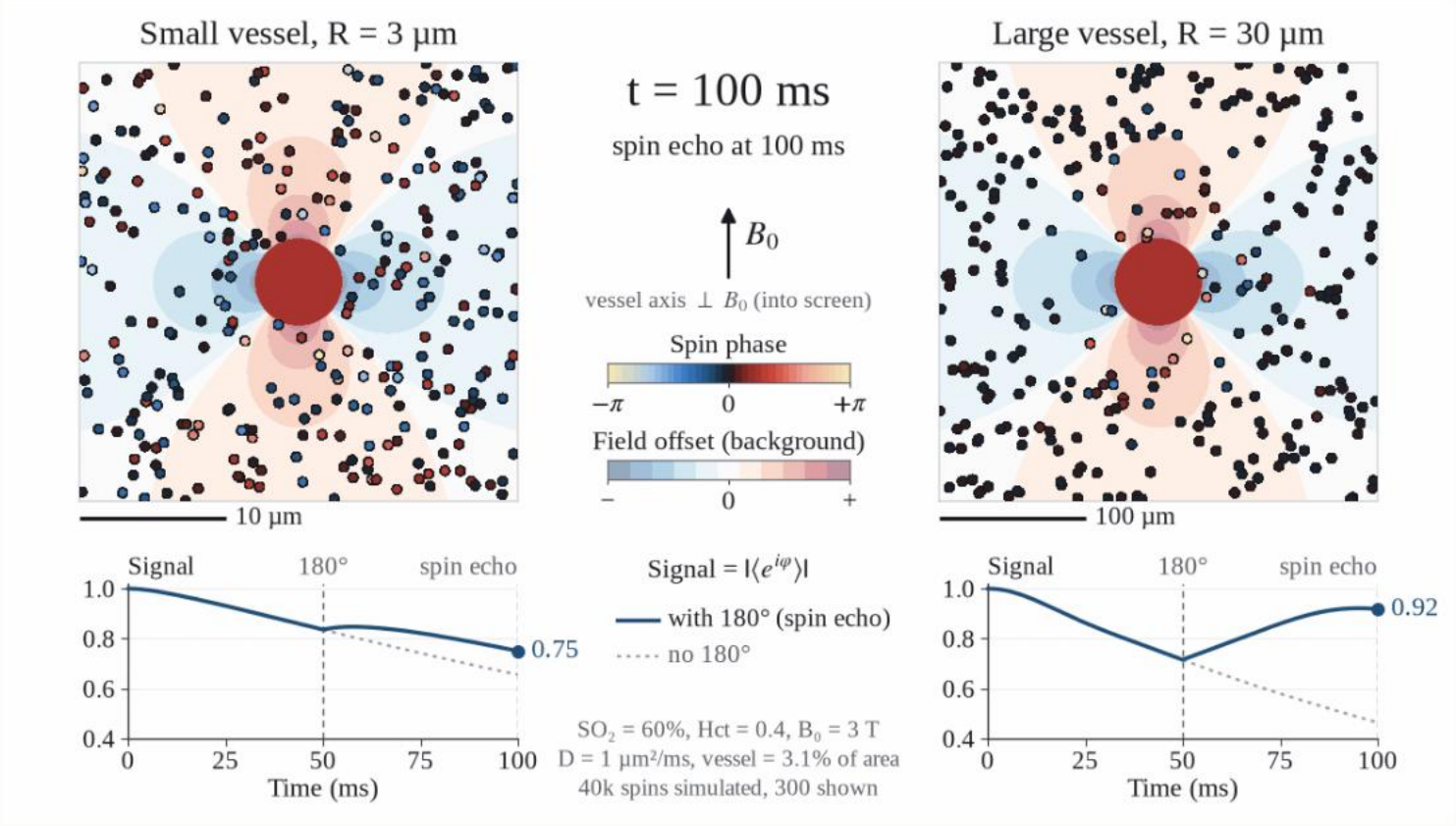
B The 180° at 50 ms starts reversing static dephasing, refocused at the spin echo at 100 ms. T2 decay continues, so the signal still falls, more slowly.

C After the spin echo, dephasing builds again.

The slopes of A and B separate R2 from R2′. WM, GM and CSF already differ.

Lin, Lu & Fan, Magn Reson Med 2025, Fig. 3A (adapted from Ni et al., Magn Reson Med 2015)

Why one curve can report SO_2, CBV and R



SO_2

Less oxygen means more paramagnetic deoxyhemoglobin and a stronger field around each vessel.

CBV

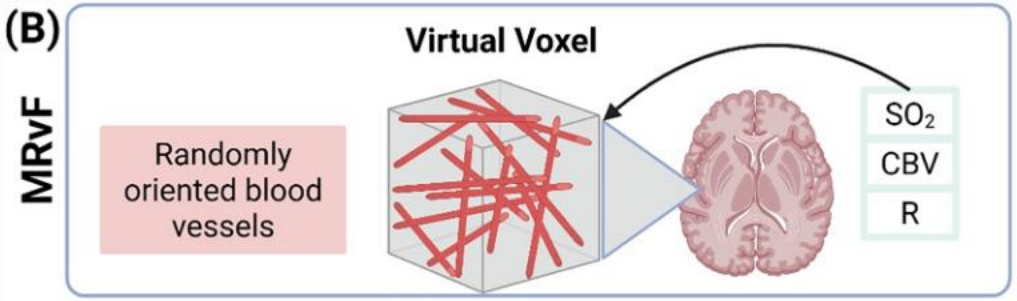
More vessels put more of the voxel inside those fields.

R

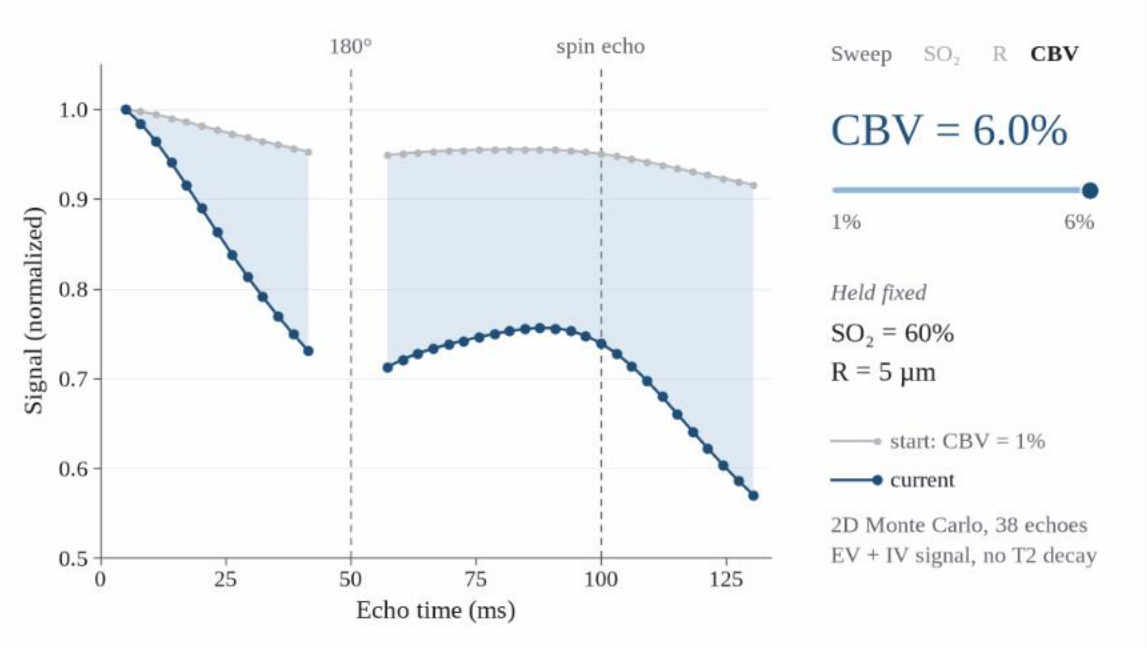
Near small vessels, water diffuses through the field, so the 180° only partly undoes the dephasing. Near large vessels the field a spin sees barely changes, so the spin echo recovers.

Based on Lin et al. Section 3. Animation: one vessel perpendicular to B0 at 3 T, D = 1 µm²/ms, T2 decay off, simulated for this talk

MRvF dictionaries come from virtual voxels

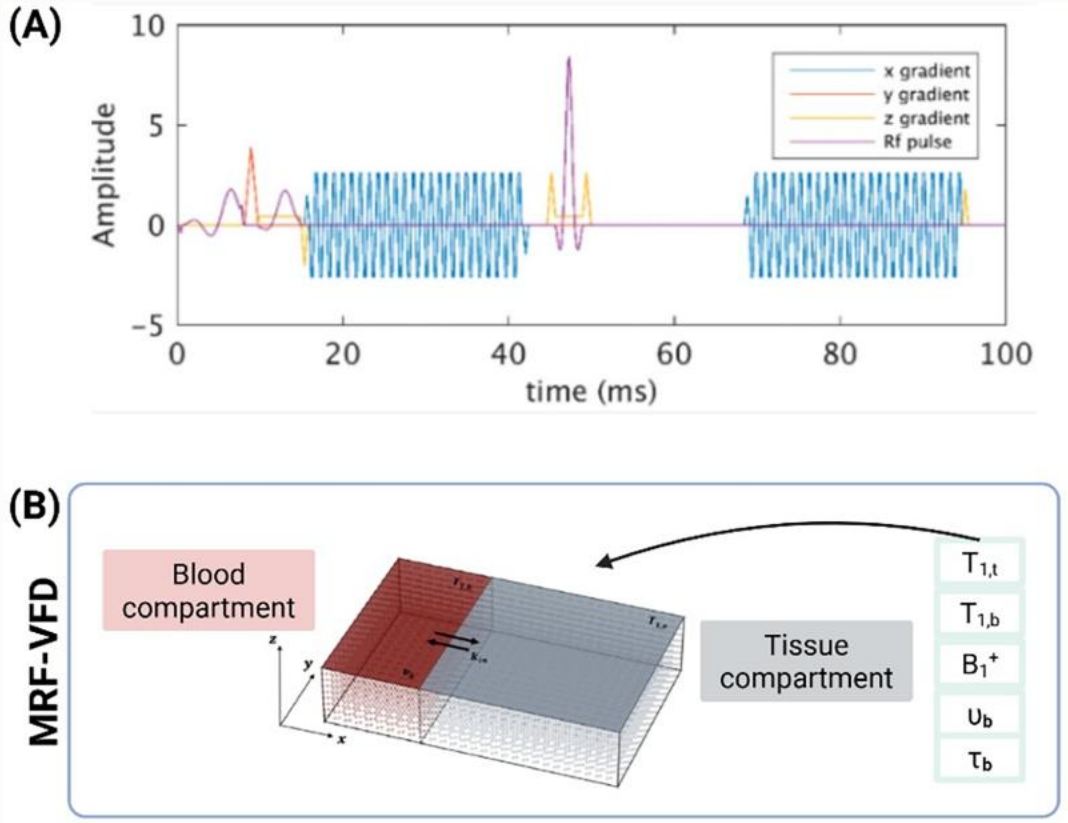


- Monte Carlo: random cylinders and diffusing water give the GE and SE signal for each SO_2, CBV and R
- Christen 2014: 52,920 entries over SO_2 0–100%, CBV 0.5–25%, R 1–25 µm, with scans before and after ferumoxytol
- No contrast: SAGE at under 5 s per map, fast enough for CVR (Wheeler); bSSFP (Coudert)
- In rodent tumors, realistic 3D vessel networks matched histology better than cylinders (Delphin)



Simulated for this talk (T2 decay off): the 38-echo GESFIDE signal as SO_2, then R, then CBV change

MRF-VFD: exchange across the blood–brain barrier



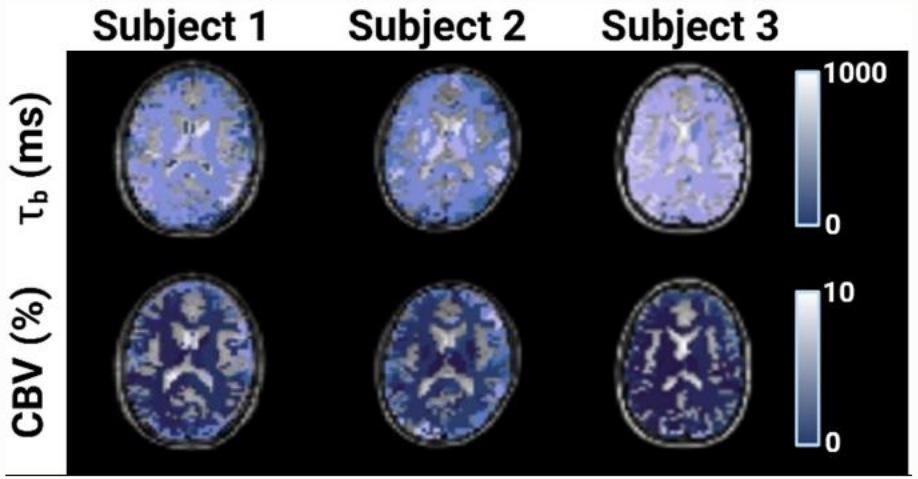


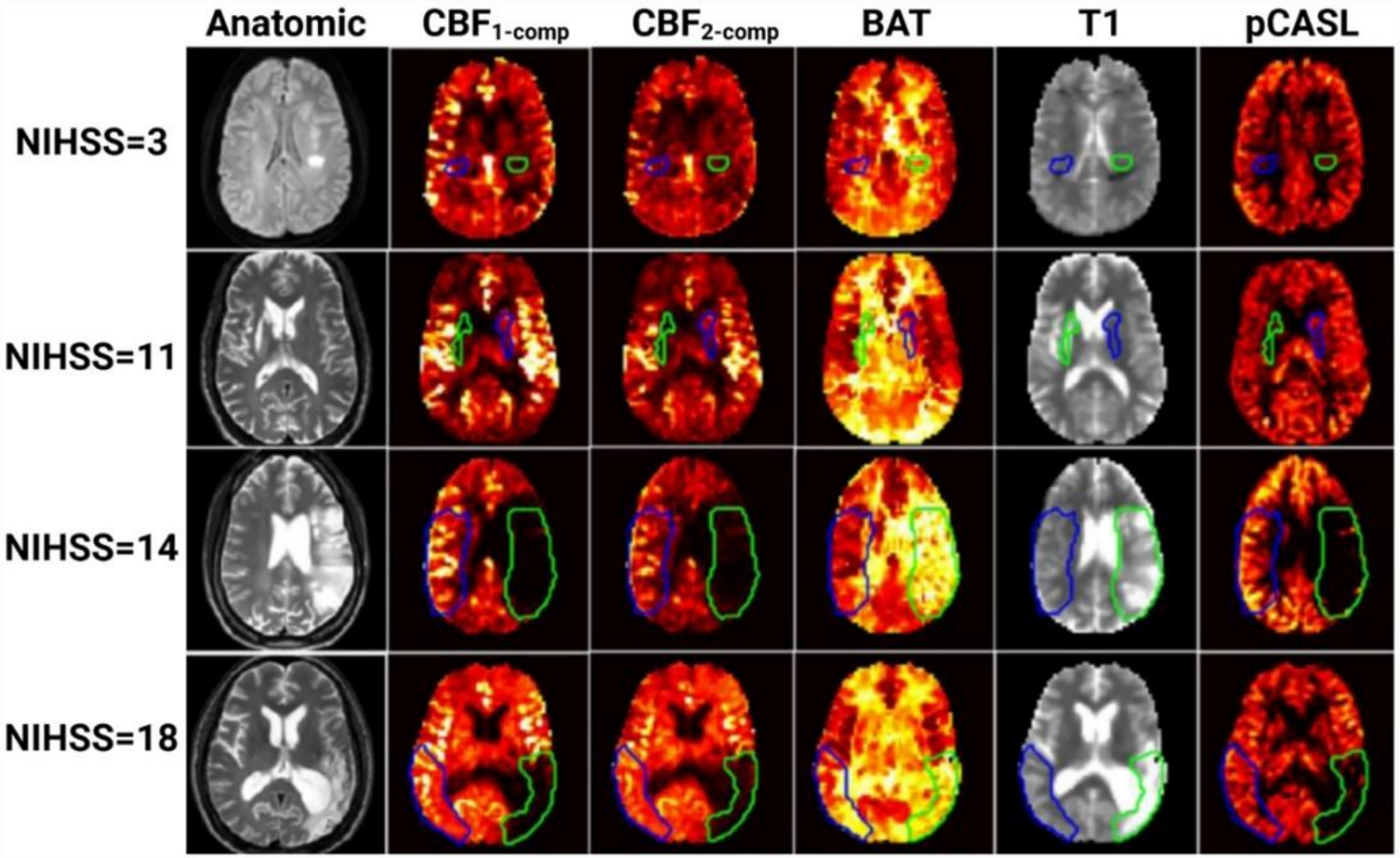
Fig. 5 (two of five rows): non-contrast MRF in three healthy volunteers (Thomson 2024)

τb: how long water stays in a vessel before it crosses the barrier.

Gray matter CBV is higher than white matter; τb and blood T1 look similar across regions.

With contrast, DCE-MRF cut the temporal variability of T1 two- to three-fold vs DCE-MRI (MacAskill; mice with breast tumors).

Clinical use: stroke and Moyamoya disease



Ischemic stroke, 34 patients (H. Fan et al. 2022)

	Lesion	Normal
CBF, 1-compartment (mL/100 g/min)	15.1	28.6
BAT (ms)	1500	900

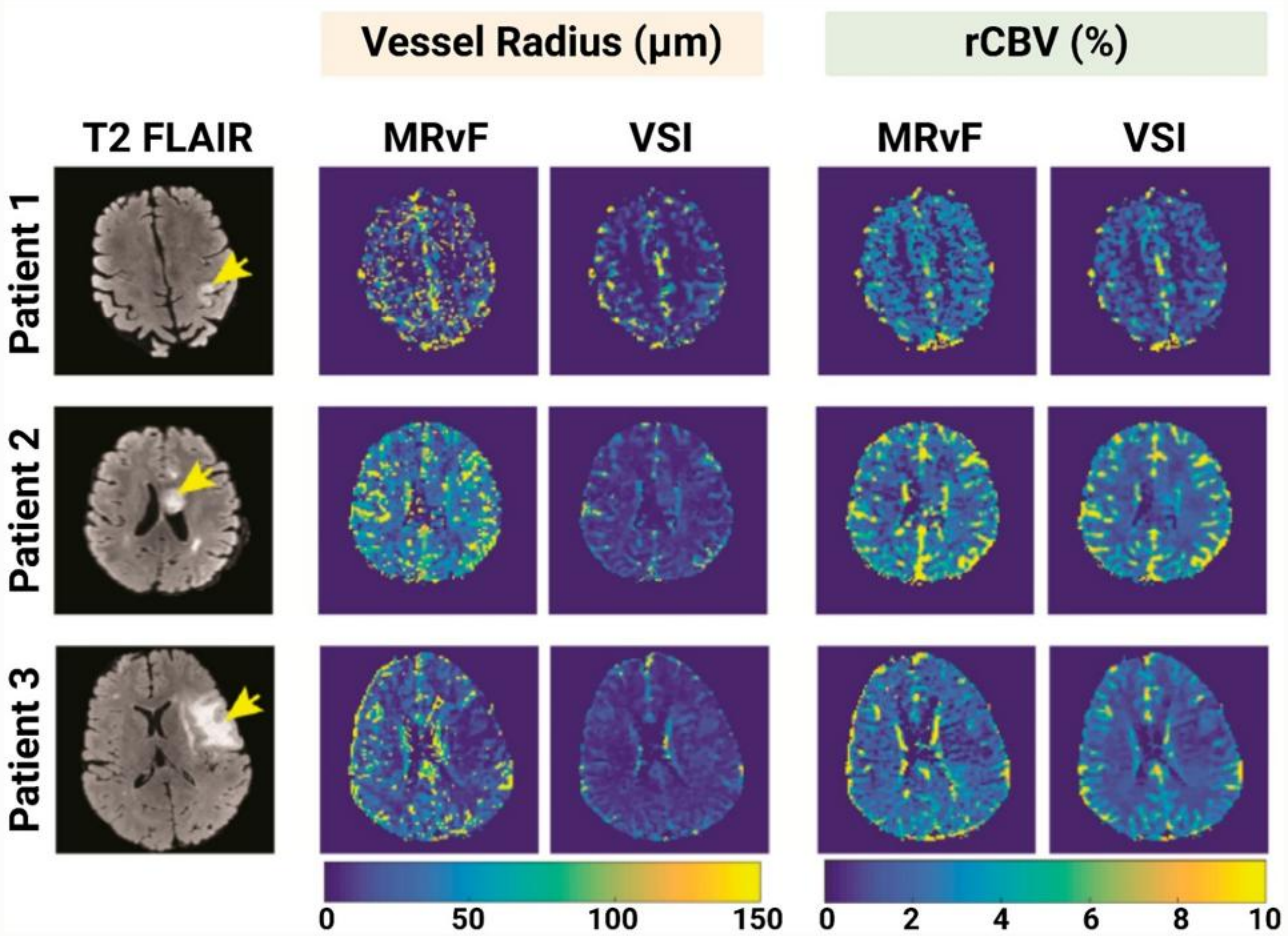
Values tracked stroke severity (NIHSS, mRS), and lesions were classified better than with pCASL.

Moyamoya, 21 patients and 16 controls (Su 2022)

Longer BAT behind narrowed arteries; CBF preserved there and raised in the posterior circulation, read as collateral flow.

Lin, Lu & Fan, Magn Reson Med 2025, Fig. 6 (adapted from H. Fan et al., Stroke 2022), Table 3 and Section 5.1

Clinical use: tumors and vessel remodeling



Gliomas, 6 patients (Venugopal 2023)

MRvF radius and rCBV maps agree with vessel size imaging; the authors report better robustness to noise and higher resolution.

Rat gliomas (Lemasson 2016)

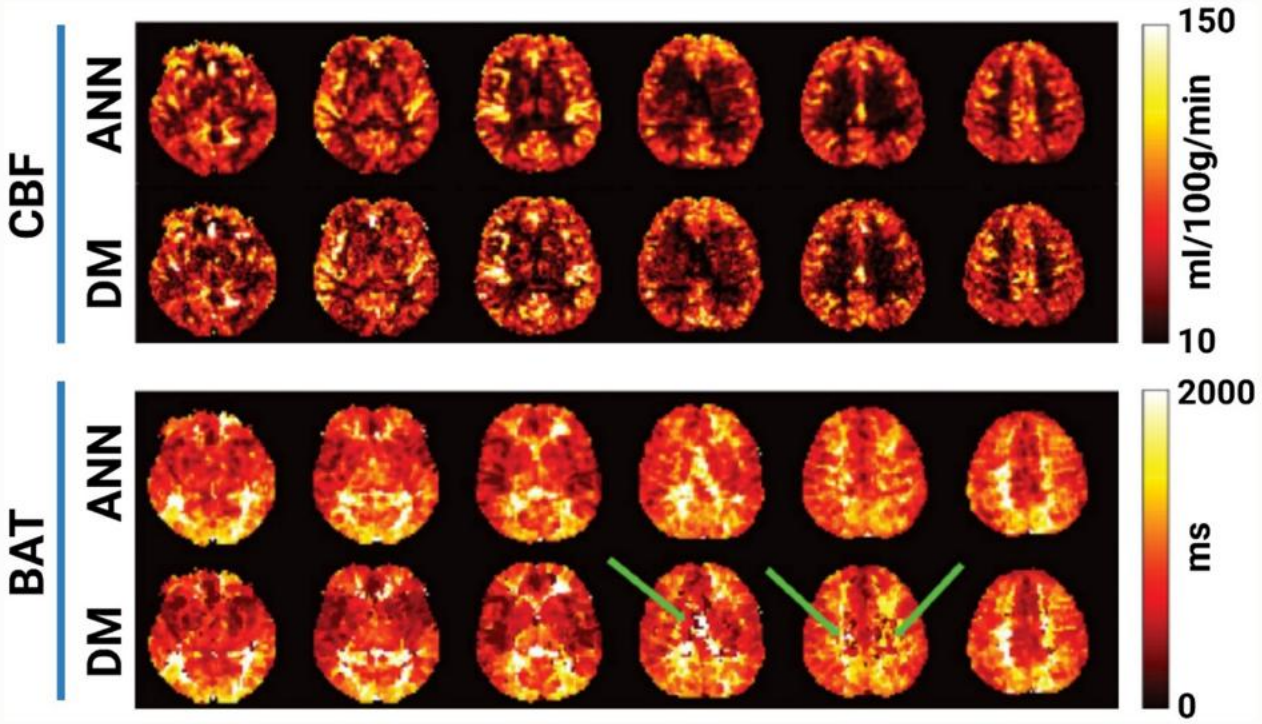
SO_2 was about 30% in 9L tumors vs 45–50% in C6 and F98. qBOLD put 9L at 40–50%.

Atherosclerotic mice (Pouliot 2017)

Larger vessels (9.2 vs 7.4 μm), more CBV (8.6 vs 6.9%) and lower SO_2 (48 vs 62%) than wild type.

Lin, Lu & Fan, Magn Reson Med 2025, Fig. 7 (adapted from Venugopal et al., Cancers 2023), Table 3 and Section 5

Machine learning replaces the dictionary search



Caveat: most networks learn from simulations (this one was trained on experimental data). A voxel unlike the training data can still return a plausible but wrong value, with no warning.

Time	Dictionary matching	Network
MRF-ASL maps, total (H. Fan et al.)	3 h 12 min	3.6 s
Per voxel (DeepMARS, Zhang 2020)	> 4 s	< 0.5 ms

For MRvF

A network trained on simulated GESFIDE dictionaries was more accurate than matching for CBV, R and SO_2 (Lin et al., ISMRM 2024, abstract 3569). MARVEL uses an LSTM trained on 3D vessel simulations.

Lin, Lu & Fan, Magn Reson Med 2025, Fig. 8 (adapted from H. Fan et al., Magn Reson Med 2021) and Section 6.2

What still limits physiological MRF

1. Field inhomogeneity. B0 and B1+ errors bias the fit. Map them and add them to the dictionary, use sequences less sensitive to B0 (FISP, FLASH rather than bSSFP), or correct with deep learning.

2. Signal-to-noise. MRvF needs an SNR of about 20 or more to match well (Wheeler). In ASL the label is only about 1% of the signal.

3. Model assumptions. Dictionaries are usually built for healthy tissue and simplified models. In disease the real signal can sit outside every entry, which biases the match.

4. Validation and standards. Static phantoms, then flow phantoms, then in vivo against PET. Sites need shared protocols (HARP, QIBA).

Directions the review points to: brain water exchange and glymphatic flow with multi-echo or long-TE ASL, and simultaneous PET/MR.

My take, and questions for discussion

What I found most useful

One framework for ASL, vascular and fluid-dynamics MRF: acquisition, dictionary, matching.

Table 2 works as a lookup: sequence, dictionary ranges, outputs and matching method for each study.

It is direct about the limits: SNR, field bias, and dictionaries built for healthy tissue.

Questions I have

1. Besides PET, what does the lab use as a reference for vessel radius and SO_2 in humans?
2. The dictionary depends on how it was simulated (BOLDsωimsuite shows methods differ in radius error). Should dictionaries be benchmarked and versioned like sequences?
3. A network inherits its dictionary's assumptions. Besides physics-informed training, how would we flag voxels outside its range, such as tumor or infarct?

How this connects to my rotation

BOLDsωimsuite

Vessel geometry, field offsets, diffusion, signal

→

GESFIDE dictionary

The lab's 38-echo timing, over SO_2, CBV and R

→

Estimation

Dictionary matching vs a neural network

→

Lab data

CBV, vessel radius and SO_2 maps

Questions I want to answer

Is a network as accurate as dictionary matching for CBV, R and SO_2?

How much faster is it?

How does it behave at low SNR and outside its training range?

First steps

Run Jack's BOLDsωimsuite notebook end to end

Reproduce the reference GESFIDE simulation

Rebuild my test dictionary in BOLDsωimsuite and test matching on noisy signals

Thank you

Questions and discussion

Lin C-T, Lu H, Fan AP. Technical review of magnetic resonance fingerprinting applications in cerebral physiology. *Magn Reson Med*. 2025. doi:10.1002/mrm.70216

Andres Michel · Fan Lab rotation · October 2026